Experimental Study on NBTI Degradation Behaviors in Si pMOSFETs Under Compressive and Tensile Strains

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***Abstract*— In this letter, we experimentally investigated the effects of four types of strains, uniaxial tensile strain, uniaxial compressive strain, biaxial tensile strain, and biaxial compressive strain, on the negative bias temperature instability (NBTI) degradation behaviors of Si pMOSFETs. The strains were applied using a wafer bending system to avoid processing effects on the NBTI characteristics as a result of strain engineering. We confirm experimentally, for the first time, that both uniaxial and biaxial compressive strains are advantageous in terms of the NBTI improvement in Si pMOSFETs. However, the NBTI reliability was degraded under both uniaxial and biaxial tensile strains. These results could not be explained by considering only the gate leakage current change due to the strain. The strain-induced modulation of the interaction between the carriers and Si–H bonds at the interface must also be considered.**

***Index Terms*— Strained Si, pMOSFETs, all types of strains, negative bias temperature instability (NBTI).**

I. Introduction

STRAIN technology has been widely used to improve the performances of Si CMOS devices [1]–[3]. It would be useful to clarify the mechanisms that affect the strained device's reliability behaviors, which have shown differences when compared with unstrained devices [4]–[7]. For p-channel MOSFETs, the primary reliability concern is the negative bias temperature instability (NBTI) [8]. Together with the increases in performance introduced by the strain technology, the NBTI behaviors in the strained pMOSFETs continue to receive increasing attention. NBTI has been reported to increase with biaxial tensile strain [9], whereas uniaxially strained pMOSFETs with SiGe S/D regions exhibit improved NBTI reliability [10]. However, different stain processing methods were used in these previous works. Therefore, a universal understanding of the effect of strain on NBTI is not yet available despite its critical importance.

Process-induced strain utilizes different steps and techniques that are not used in unstrained devices, thus making it difficult to identify the physical mechanisms related to strain that affect the NBTI behavior. For instance, Ge atoms in the channel and in the SiGe source and drain [5] or excess H atoms in the stressor SiN films would also greatly affect NBTI reliability, making it difficult to compare strained devices with unstrained devices and preventing an understanding of the underlying physical mechanism [9].

In this letter, strains were applied to the devices by using a wafer bending system that avoids the aforementioned difficulties. Using this system, we studied the effects of different types of strain, including uniaxial tensile strain, uniaxial compressive strain, biaxial tensile strain and biaxial compressive strain, on the NBTI behavior. The NBTI performance was studied under both constant gate voltage (V_g) and constant gate current (I_g). For the first time, we experimentally demonstrate that both uniaxial and biaxial compressive strains can improve the NBTI reliability of Si pMOSFETs, whereas both uniaxial and biaxial tensile strains can degrade the NBTI reliability.

II. Experiment

All of the devices measured in this letter were fabricated from bulk Si with a standard CMOS process. The substrate concentration was 3×10^{17} cm^{-3}. Gate oxides were composed of pure SiO_2 grown at an oxidation temperature of 850 °C with a thickness of approximately 8 nm. Doped poly-Si was used as the gate electrode. The pMOSFETs used in this letter had a gate length of 100 μm and width of 100 μm. Uniaxial and biaxial strain (tensile and compressive) can be applied by using specific top and bottom plates, and the amount of applied stress was adjusted and calculated according to the relative displacement of the top and bottom plates (Fig. 1). The threshold voltage (V_{th}) was obtained through the linear

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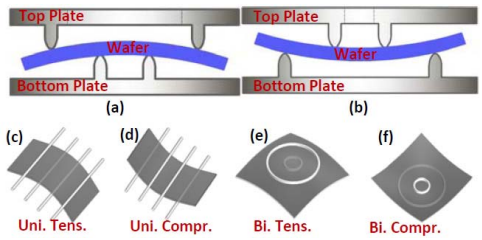

Fig. 1. Illustration of the wafer bending system used to apply (a) tensile stress and (b) compressive stress. Diagram of the specific top and bottom plates used to apply (c) uniaxial tensile strain, (d) uniaxial compressive strain, (e) biaxial tensile strain and (f) biaxial compressive strain.

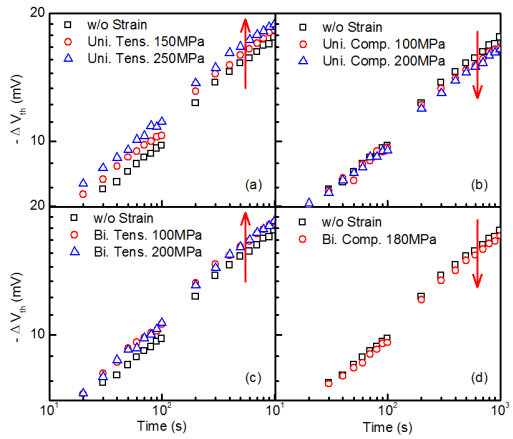

Fig. 2. Time evolution of the ΔV_{th} of pMOSFETs under (a) uniaxial tensile strain, (b) uniaxial compressive strain, (c) biaxial tensile strain and (d) biaxial compressive strain during the NBTI stress of a constant gate oxide electric field (E_{ox}) of 9 MV/cm at room temperature.

extrapolation method to surpass the effect of strain-induced varied mobility and gate-to-channel capacitance (C_{gc}) [11]. V_{th} shift (ΔV_{th}) measurements were carried out by the stress-measure-stress method. The V_g step for $I_d - V_g$ measurements was 10 mV.

III. Results and Discussion

A gate oxide electric field (E_{ox}) of 9 MV/cm was used for the NBTI stress. As illustrated in Fig. 2(a), pMOSFETs under uniaxial tensile strains have larger ΔV_{th} values compared with unstrained devices, and the difference in V_{th} increases with increasing strain. By contrast, pMOSFETs under uniaxial compressive strain result in a smaller NBTI-induced ΔV_{th} as a function of the applied strain [Fig. 2(b)]. Similarly, ΔV_{th} increases under biaxial tensile strains [Fig. 2(c)], consistent with the result in [9], and decreases under biaxial compressive strains [Fig. 2(d)]. In summary, the NBTI reliability of pMOSFETs increases under compressive strains (uniaxial and biaxial) but is degraded under tensile strains (uniaxial and biaxial) during both the stress and relaxation stages [Fig. 3(a), (b)]. Although the difference in the ΔV_{th} discussed here is several mV, it does not prevent us from drawing conclusions because the device-to-device variability of ΔV_{th} after the NBTI stress is considerably smaller than 1 mV (data not shown).

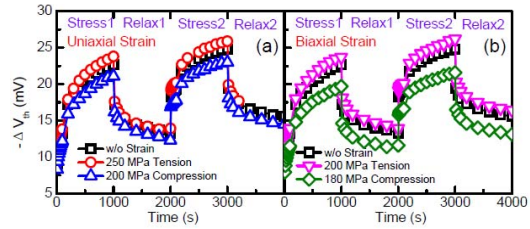

Fig. 3. Time evolution of the ΔV_{th} of pMOSFETs under (a) uniaxial strain and (b) biaxial strain during two cycles of interrupted NBTI stress (stress 1,000 s, relaxation 1,000 s, $E_{ox} = 9$ MV/cm, T = 25 °C).

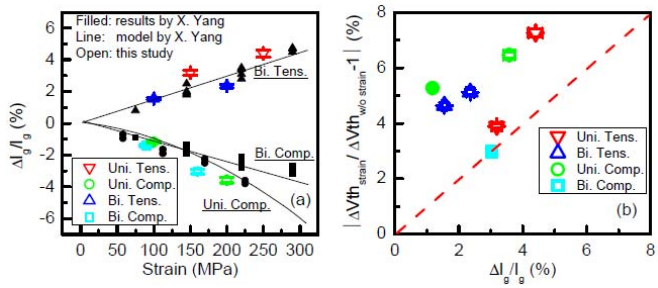

Fig. 4. (a) The change in I_g versus the applied strain under four types of strains. The filled symbols and lines show the experimental results and theoretical model in [11]. The open symbols are the results obtained in this letter. (b) The change in ΔV_{th} between devices with and without strain versus the change in I_g under different strains.

During NBTI stress, ΔV_{th} is strongly correlated with I_g as follows:

$$\Delta V_{th}(t) = \alpha . I_g . In(\frac{t}{\tau_0}) \quad (1)$$

where t is the stress time and α and τ_0 are constants that are process dependent [13]. I_g for each NBTI strain condition was monitored and compared with the results in [12] [Fig. 4(a)]. The decrease (increase) in I_g for compressive (tensile) strains primarily results from the repopulation of holes into a sub-band with a larger (smaller) out-of-plane effective mass (m_z). However, if we assume that the mechanical strain could not change α and τ_0, the changes of ΔV_{th} of the strained Si devices are still larger than expected according to Equation (1) [Fig. (4b)]. After a constant I_g stress of -250 nA/cm^2, the strain-modulated NBTI behaviors under different types of strains still exist and are similar to those observed under constant E_{ox} stress, although the difference is considerably smaller (Fig. 5). On the other hand, the substrate current (I_{sub}) is also constant during the NBTI stress process. Thus, merely the strain-induced modulation of I_g cannot explain the NBTI behavior of strained pMOSFETs.

As discussed in [9], because strain modulates the population of holes in the light- and heavy-hole sub-bands, it can also affect the interactions between holes and Si-H bonds at the interface because the light- and heavy-hole sub-bands have different distribution functions. In this case, the inversion layer capacitance (C_{inv}) and thus C_{gc} would also be tuned by the strains. In fact, C_{gc} increases with increasing compressive strain but decreases with increasing tensile strain, as shown in Fig. 6. If we assume that the capacitance of the gate

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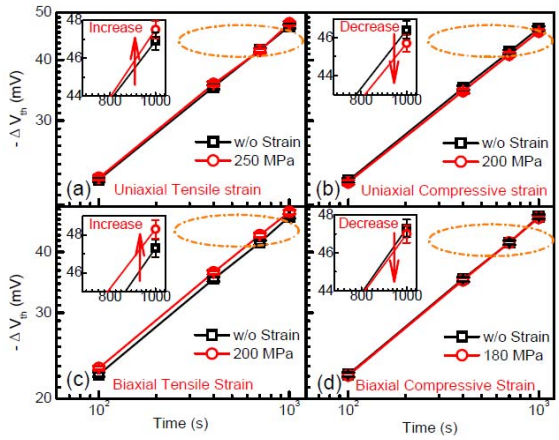

Fig. 5. Time evolution of the ΔV_{th} of pMOSFETs under (a) uniaxial tensile strain, (b) uniaxial compressive strain, (c) biaxial tensile strain and (d) biaxial compressive strain during constant NBTI stress of I_g of -250 nA/cm^{-2} at room temperature.

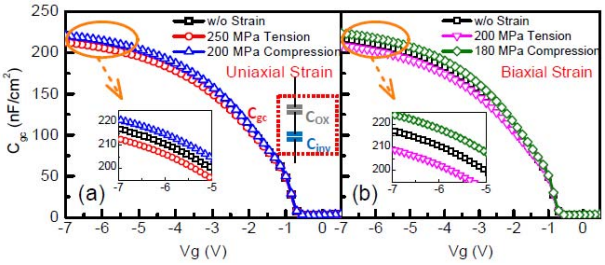

Fig. 6. C_{gc} versus V_g of pMOSFETs under (a) uniaxial strain and (b) biaxial strain.

oxide (C_{ox}) is independent of the strain conditions in the compressive strained channel, C_{inv} is larger and the average inversion layer width is smaller compared with the unstrained device. This result is consistent with strain-induced holes repopulating into the heavy-hole sub-band (Fig. 7), which has a smaller average-inversion-layer width [12].

As a result, there is less wave function penetration into SiO_2, resulting in a weaker interaction between holes and the Si-H bonds at the interfaces in compressive strained devices (Fig. 7). Finally, compressive strained devices display improved NBTI reliability. A similar explanation could be applicable to the inferior NBTI reliability observed in tensile-strained devices. Therefore, the observed NBTI experimental results are attributable to the I_g change and strain-induced change in the interaction between holes and Si-H bonds at the MOS interface. However, we still cannot exclude other possible mechanisms, such as the change in the trapping time constant, τ_0, due to strain. Therefore, further study is still needed to clarify the reason for the different NBTI behaviors of unstrained and strained devices.

Finally, considering the repopulation of holes and out-of-plane effective mass variation, no essential difference would occur even when the strain increases to several GPa. The conclusion drawn in this letter is still a reference for the real strained devices used in industry, which have considerably higher strains.

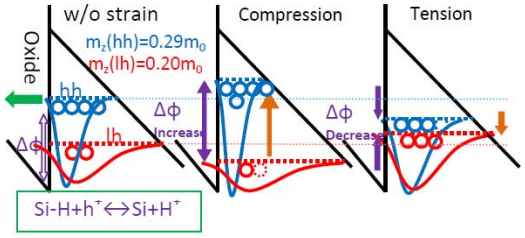

Fig. 7. Schematic showing the strain-induced band structure modulation, hole repopulation and the effects of strain on NBTI.

IV. Conclusion

We have comprehensively studied the effects of all four types of strains on the NBTI reliability behaviors of Si pMOSFETs. It has been experimentally confirmed, for the first time, that both uniaxial and biaxial compressive strains could improve the NBTI reliability of Si pMOSFETs. By contrast, the NBTI reliability is degraded in both uniaxial and biaxial tensile-strained Si pMOSFETs. These observations can be explained by considering the change in the interaction between the carriers and Si-H bonds at the interface and the gate current through the interface.

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